ECOR2050: Design and Analysis of Engineering Experiments

Assignment 3: Due November 4, 2026

Carleton UNIVERSITY
Canada's Capital University

Instructions:

1. Solutions should be compiled in a single file and submitted to Brightspace.
2. Answer all parts of each question.
3. Show all manual equations, calculations, and units.
4. Plots must be labelled, clear, and legible.
5. Plots must be completed in software.
6. Every solution must include a concluding statement that interprets the numeric result. Correct numeric results without a concluding statement will lose marks.

Q1. A solar panel manufacturer claims the average power output of their panels under standard testing conditions is 250 W. To investigate this claim, a sample of 15 randomly selected panels from the manufacturer are tested under these conditions. The sample mean and standard deviation are 246 W and 12 W, respectively.

(a) Determine the probability of obtaining a sample mean as low or lower than 246 W, assuming the actual mean is 250 W. Show the result to two significant figures. Include any necessary assumptions. **[4 marks]**

(b) What does this answer imply about the actual mean being 250 W? **[1 mark]**

Q2. A materials group is investigating the shear strength of an alloy when annealing it at two different temperatures (low and high). The shear strength is tested on random samples of the material after each annealing process, with the results shown in the table below (in units of MPa):

Temp.	Shear strength (MPa)													
Low	196	203	206	222	206	217	184	205	199	190	215	212	209	193
High	206	202	204	202	193	202	199	198	197	201				

(a) Find the 90% confidence interval for the difference between the shear strength of the two alloy samples. Include any necessary assumptions. **[6 marks]**

(b) What does the confidence interval reveal about the strengths of the alloys produced under each process? Annealing at which temperature helped produce the stronger alloy? **[2 marks]**

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Q3. During the 1999 and 2000 baseball seasons, there was speculation that the unusually large number of home runs was due in part to a livelier ball. One way to test "liveliness" is to launch the ball at a vertical surface at a known initial velocity and measure its outgoing velocity. Taking the ratio of these two speeds give a coefficient of restitution, $R = v_o/v_i$. The following table contains data on R from 41 randomly selected baseballs launched from a pitching machine at the same speed toward the same oak surface.

0.6248	0.6237	0.6218	0.6159	0.6298	0.6192	0.6520	0.6368
0.6220	0.6151	0.6121	0.6348	0.6226	0.6280	0.6096	0.6100
0.6107	0.6392	0.6230	0.6131	0.6223	0.6497	0.6435	0.5978
0.6351	0.6275	0.6261	0.6262	0.6262	0.6314	0.6128	0.6403
0.6121	0.6049	0.6170	0.6134	0.6310	0.6065	0.6514	0.6141
0.6318							

(a) Is there evidence to support the assumption that the coefficient of restitution measured here is normally distributed? **[2 marks]**

(b) Find a 99% confidence interval on the mean coefficient of restitution. **[3 marks]**

(c) From a large study on the same baseballs, the reported variance of the population coefficient of restitution is $\sigma^2 = 0.000151$. Recompute the 99% confidence interval on the mean using this value. **[2 marks]**

(d) Find an interval that contains with 99% confidence, the coefficient of restitution measured for a single ball randomly sampled from these years. **[2 marks]**

(e) Explain the difference between the intervals computed in (b), (c), and (d). **[2 marks]**